HCML Monitoring Portal: Physical Data Model (PDM), MySQL 8.0

Hosts & inventory

hosts

bigint	hostid	PK	
varchar(128)	host		technical name
varchar(128)	name		visible name
int	status		0 monitored, 1 not monitored, 3 template
int	maintenance_status		1 in maintenance

hstgrp

bigint	groupid	PK	
varchar(255)	name		
int	type		0 host group, 1 template group

hostid, cascade

host_inventory

bigint	hostid	PK,FK	no row when inventory is disabled
varchar(128)	site_city		
text	location		
varchar(128)	poc_1_name		
varchar(128)	poc_1_email		

hostid, cascade

interface

bigint	interfaceid	PK	
bigint	hostid	FK	
int	main		1 the default interface
int	type		1 agent, 2 SNMP, 3 IPMI, 4 JMX
varchar(64)	ip		
int	available		0 unknown, 1 up, 2 down

hostid, cascade

hosts_groups

bigint	hostgroupid	PK
bigint	hostid	FK
bigint	groupid	FK

groupid, cascade

hostid

host_tag

bigint	hosttagid	PK	
bigint	hostid	FK	
varchar(255)	tag		site, owner, criticality
varchar(255)	value		

hostid, cascade

interfaceid

Items & stored values

items

bigint	itemid	PK	
bigint	hostid	FK	
bigint	interfaceid	FK	
varchar(255)	name		
varchar(2048)	key_		
int	value_type		picks the history and trend table
varchar(255)	units		
int	status		0 enabled, 1 disabled

itemid, cascade

item_rtdata

bigint	itemid	PK,FK	
int	state		0 normal, 1 not supported
varchar(2048)	error		

itemid

item_tag

bigint	itemtagid	PK	
bigint	itemid	FK	
varchar(255)	tag		link_group, link_role
varchar(255)	value		

itemid, value_type 0, no FK

history

bigint	itemid	PK	no FK to items
int	clock	PK	
int	ns	PK	
double	value		

itemid, value_type 3, no FK

history_uint

bigint	itemid	PK	no FK to items
int	clock	PK	
int	ns	PK	
bigint	value		

itemid, value_type 0, no FK

trends

bigint	itemid	PK	no FK to items
int	clock	PK	start of the hour
int	num		values in that hour
double	value_avg		
double	value_max		

itemid, value_type 3, no FK

trends_uint

bigint	itemid	PK	no FK to items
int	clock	PK	start of the hour
int	num		
bigint	value_avg		
bigint	value_max		

itemid

Triggers, events & problems

events

bigint	eventid	PK	
int	source		0 triggers, 4 services
int	object		0 trigger, 4 item, 5 LLD, 6 service
bigint	objectid		no FK, object says what it points at
int	clock		
int	value		1 problem, 0 recovery
int	acknowledged		set by event.acknowledge
varchar(2048)	name		
int	severity		0 to 5

triggers

bigint	triggerid	PK	
varchar(2048)	expression		reaches items through functions
int	manual_close		1 Close permitted
varchar(255)	opdata		

eventid, cascade

acknowledges

bigint	acknowledgeid	PK	
bigint	userid	FK	the API token user
bigint	eventid	FK	
int	clock		
varchar(2048)	message		
int	action		bitmask 1 close, 2 ack, 4 message

eventid, cascade

event_recovery

bigint	eventid	PK,FK	the problem event
bigint	r_eventid	FK	the event that resolved it

eventid, cascade

objectid when object=0, no FK

problem

bigint	eventid	PK,FK	
bigint	objectid		no FK
int	clock		
bigint	r_eventid	FK	NULL while open
varchar(2048)	name		
int	acknowledged		
int	severity		

triggerid

functions

bigint	functionid	PK
bigint	itemid	FK
bigint	triggerid	FK

eventid, cascade

problem_tag

bigint	problemtagid	PK
bigint	eventid	FK
varchar(255)	tag	
varchar(255)	value	

eventid, cascade

Services & SLA

services

bigint	serviceid	PK	
varchar(128)	name		
int	status		-1 OK, else a severity
int	algorithm		
text	description		

serviceid, cascade

service_tag

bigint	servicetagid	PK
bigint	serviceid	FK
varchar(255)	tag	
varchar(255)	value	

servicedownid, cascade

serviceupid, cascade

services_links

bigint	linkid	PK	
bigint	serviceupid	FK	the parent
bigint	servicedownid	FK	the child

sla

bigint	slaid	PK	
varchar(255)	name	UK	
int	period		0 daily to 4 annually
double	slo		target percent
varchar(50)	timezone		
int	status		0 disabled, 1 enabled

slaid, cascade

serviceid, cascade

service_problem

bigint	service_problemid	PK	
bigint	eventid	FK	a problem row, not an event
bigint	serviceid	FK	
int	severity		

matched on tag and value, no key

sla_service_tag

bigint	sla_service_tagid	PK
bigint	slaid	FK
varchar(255)	tag	
int	operator	
varchar(255)	value	